Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 18)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

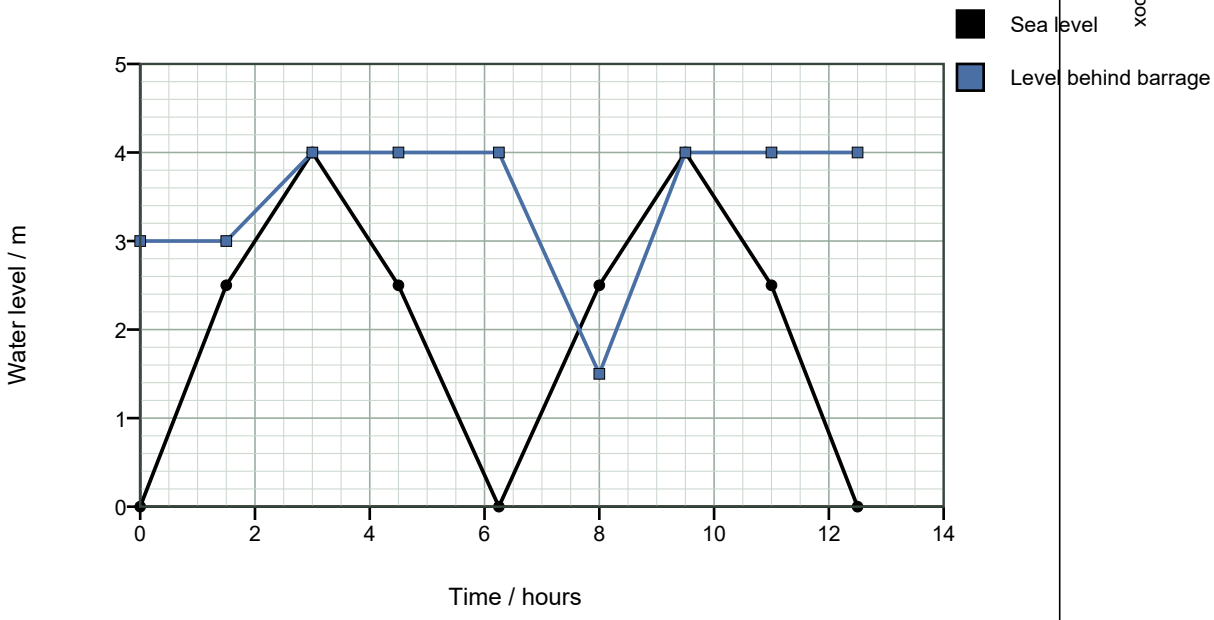
Question	Mark
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Figure 1 shows the water level inside and outside a tidal barrage over one 12.5-hour tidal cycle and the periods when electricity is generated.

Figure 1



0 1 . 1 Use Figure 1 to explain when electricity is generated by the barrage.

[2 marks]

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0 1 . 2 Explain why tidal power is intermittent but predictable.

[2 marks]

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0 1 . 3 Explain the difference between spring and neap tides and why this affects output.

[2 marks]

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0 1 . 4 Explain how a tidal lagoon can be used to generate electricity when there is no tidal flow.

[2 marks]

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0 1 . 5 Give **two** coastal features that make a site suitable for tidal power.

[2 marks]

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0 2 . 1 Which of the following statements about stratospheric ozone is correct? Tick one box.

[1 mark]

A Ozone is destroyed only by CFCs ☐

B Ozone absorbs mainly infrared radiation ☐

C Ozone is formed when monatomic oxygen reacts with diatomic oxygen ☐

D Ozone concentrations are highest at ground level ☐

0 2 . 2 Explain the difference between the roles of ozone in the stratosphere and in the troposphere.

[2 marks]

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0 2 . 3 Explain why HCFCs cause less ozone depletion than CFCs.

[2 marks]

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0 3 . 1 Explain why electronic meters must be calibrated before use and describe how a pH meter is calibrated.

[3 marks]

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0 3 . 2 Describe how wind velocity measurements with an anemometer should be standardised.

[3 marks]

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0 3 . 3 Describe how a whirling hygrometer is used to measure humidity.

[2 marks]

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0 3 . 4 Describe **two** methods of measuring nitrate concentration in water.

[2 marks]

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0 4 . 1 Which of the following water sources is usually the most convenient for large communities? Tick one box.

[1 mark]

A Desalinated seawater ☐

B River water ☐

C Rainwater catchment ☐

D Newly discovered aquifers ☐

0 4 . 2 Explain why the quality requirements for water differ between uses, using **two** named uses.

[2 marks]

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0 4 . 3 Explain why inter-basin transfers may be used and give **one** disadvantage.

[2 marks]

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0 5 Table 1 compares the energy used to produce 1 tonne of aluminium from bauxite ore with the energy used to produce 1 tonne from recycled aluminium cans.

Table 1

Source	Energy used / GJ per tonne	CO_2 emitted / tonnes per tonne of aluminium
Bauxite ore (extraction, refining, smelting)	210	12
Recycled cans (collection, sorting, remelting)	12	0.7

0 5 . 1 Calculate the percentage energy saving from recycling aluminium rather than extracting it from ore.

[2 marks]

Answer ..... %

0 5 . 2 Give **three** advantages of recycling metals other than energy saving.

[3 marks]

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0 5 . 3 Explain **four** difficulties that reduce the effectiveness of recycling schemes.

[4 marks]

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0 5 . 4 Explain what is meant by **Cradle to Cradle** design.

[1 mark]

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0 6 . 1 Explain why oil and gas trapped in shale cannot be extracted by conventional drilling and describe how hydraulic fracturing (fracking) allows extraction.

[3 marks]

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0 6 . 2 Describe **four** environmental concerns about fracking and suggest how they may be reduced.

[5 marks]

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0 6 . 3 Explain why tar sands and oil shales are not included in most estimates of oil reserves despite containing more oil than conventional crude oil.

[2 marks]

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0 7 Table 1 shows some properties of three pollutants.

Table 1

Pollutant	State at 20 °C	Solubility in water	Solubility in lipids	Environmental half-life
DDT (insecticide)	Solid	Very low	High	About 15 years in soil
Nitrate ions	Dissolved ion	Very high	Very low	Not applicable (not degraded)
Hydrogen cyanide	Gas (denser than air)	High	Low	Hours

0 7 . 1 Define the term **environmental half-life**.

[1 mark]

0 7 . 2 Use Table 1 to explain why DDT is more likely than nitrate ions to biomagnify in a food chain.

[3 marks]

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0 7 . 3 Suggest why the release of hydrogen cyanide at Bhopal, India, in 1984 caused deaths very close to the source rather than over a wide area.

[2 marks]

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0 7 . 4 Explain how the high water solubility of nitrate ions affects the area over which they cause pollution and the concentration of the pollution.

[2 marks]

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0 7 . 5 Give **two** examples of pollutants that are forms of energy rather than substances.

[2 marks]

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0 8 . 1 Complete Table 1 to explain how each method reduces soil erosion.

[5 marks]

Table 1

Method	How it reduces erosion
Terracing	
Tied ridging	
Windbreaks / hedgerows	
Mulching	
Strip cropping	

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0 8 . 2 Explain why long-term crops such as tea, coffee or orchards reduce erosion compared with annual arable crops.

[2 marks]

0 8 . 3 Describe **three** ways livestock can be managed to reduce soil erosion.

[3 marks]

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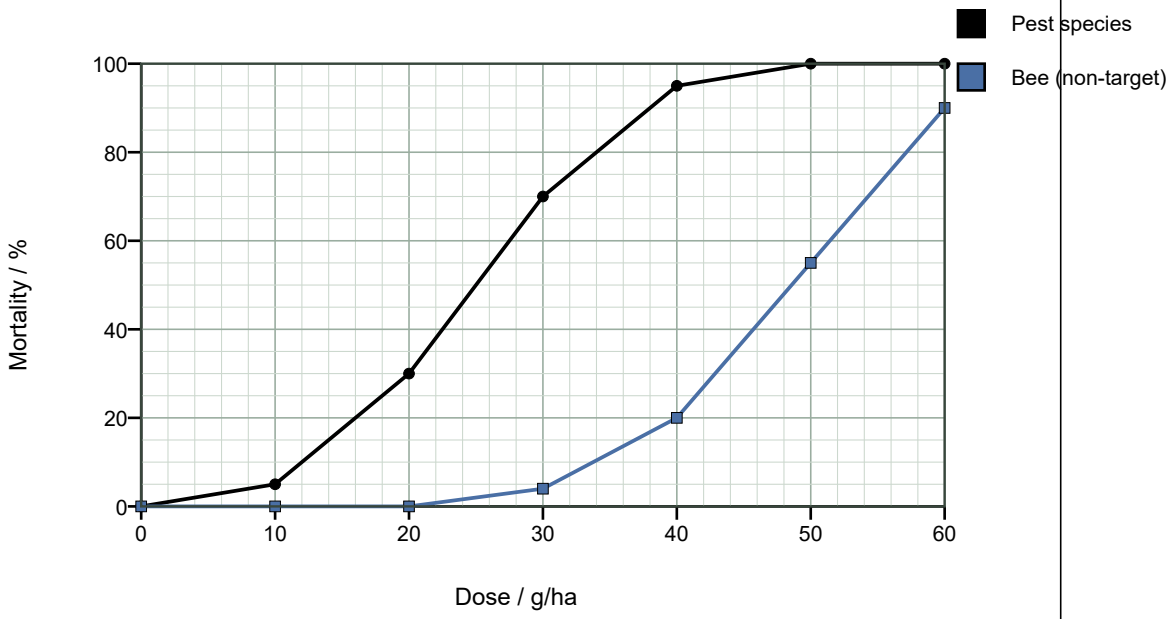
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0 9 Figure 1 shows how the percentage mortality of a pest species and of a non-target bee species change with the dose of an insecticide applied to a crop.

Figure 1



0 9 . 1 Use Figure 1 to estimate the LD_{50} of the insecticide for the pest species.

[1 mark]

Answer ..... g/ha

0 9 . 2 Define the term **MDNF** (maximum dose never fatal).

[1 mark]

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0 9 . 3 Use Figure 1 to explain why it may not be possible to kill all of the pest species without harming bees.

[3 marks]

0 9 . 4 Suggest how the farmer could reduce the harm to bees without reducing the dose applied.

[2 marks]

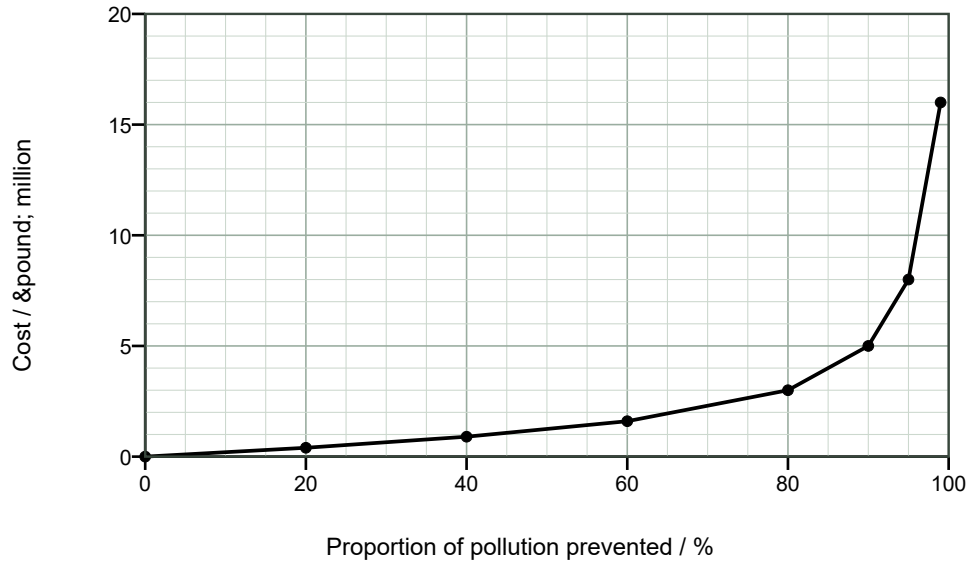
0 9 . 5 Explain why a pesticide with high specificity is less likely to cause environmental damage.

[3 marks]

10

1 0 Figure 1 shows how the cost of controlling emissions from a factory changes with the proportion of the pollutant that is prevented from being released.

Figure 1



1 0 . 1 Use Figure 1 to calculate the extra cost of increasing control from 90% to 99% of the pollutant.

[1 mark]

Answer £ million

1 0 . 2 Explain how the terms **ALARA** and **BATNEEC** are used when deciding how much pollution control is required.

[3 marks]

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1 0 . 3 Explain why the control methods that were originally adequate for a factory may no longer be sufficient.

[2 marks]

1 0 . 4 Evaluate the principles and general strategies that can be used to decide how pollution should be controlled.

[9 marks]

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Extra space

15

Write an essay on one of the following topics.

1 1 . 1 Discuss the difficulties in monitoring and predicting global climate change and evaluate the methods used to overcome them.

[25 marks]

OR

1 1 . 2 Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS